

Kellogg's® Math In Action Video Series – Cereal Nutrition Video – Student Activity Guide

With your class, watch the Kellogg's® Cereal Nutrition Video pausing (when guided to do so) and discussing your answers with your group. Be prepared to share out your answers.



1. Given the individual ingredient values, calculate out the total macro and micronutrients. Then, with the total numbers, calculate out what those values would be on a 40g serving size (complete this on your Yummy Oaties Recipe Worksheet).
2. The FDA utilizes specific rounding rules for numbers with decimals. Using the rounding rules given by your teacher, establish what the rounded nutrients will be. (complete this on your Yummy Oaties Recipe Worksheet).
3. The nutrition facts panel utilizes Daily Values (DV) percentages to help consumers make informed choices. Using the DVs below, calculate the DV percentages.
 - Total Fat 78g
 - Saturated Fat 20g
 - Cholesterol 300mg
 - Sodium 2300mg
 - Total Carbohydrate 275g
 - Dietary Fiber 28g
 - Added Sugars 50g
 - Protein 50g
 - Vitamin D 20mcg
 - Calcium 1300mg
 - Iron 18mg
 - Potassium 4700mg
4. Take your calculated values and place into the blank nutrition facts panel. Note that trans-fat and total sugars will not have a DV % due to no recommended guidance. Protein will also remain blank due to a different way to calculate DV%.
5. Find the total calories by utilizing the rounding rules. If total calories <5 , 0 is declared on the label. If 5-50, it is expressed to the nearest 5 increment and if >50 , it is expressed to the nearest 10 increment. Calculate Total Calories using the information below:
 - Fat = 9 calories/g
 - Carbohydrates = 4 calories/g
 - Protein = 4 calories/g

Nutrition Facts

Serving Size

1 cup (40g/1.4oz)

Amounts Per Serving

Calories

% Daily Value*

Total Fat ___ %

Saturated Fat ___ %

Trans Fat ___

Cholesterol ___ %

Sodium ___ %

Total Carbohydrate ___ %

Dietary Fiber ___ %

Total Sugars ___

Includes ___ Added Sugars %

Protein ___

Vitamin D ___ % • Potassium ___ %

Calcium ___ % • Iron ___ %

* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower based on your calorie needs.

Tolerance Question:

6. A supplier wants to switch out the kind of natural flavor because of a shortage. The new natural flavor has 6g of sugar/100g and 6g added sugar/100g. Given these values, how will the new flavor impact the overall recipe and label? Will the recipe be within 110% of the label?



Ingredients (per 100g)	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Total Carbs (g)	Dietary Fiber (g)	Total Sugar (g)	Added Sugar (g)	Protein (g)	Vitamin D (mcg)	Calcium (mg)	Iron (mg)	Potassium (mg)
Whole Grain Oat Flour	2	0.5	0	0	1	28	4	0	0	7	0	22	3	185
Corn Flour	2	0	0	0	0	19	2	0	0	3	0	4.5	0.5	16
Dried Strawberries	0	0	0	0	1	4	4	6	0	0	0	6.5	1	0
Sugar	0	0	0	0	0	0	0	18	18	0	0	0	0	0
Natural Flavor	0	0	0	0	2	2	0	3	3	0	0	0	0	90
Salt	0	0	0	0	3	0	0	0	0	0	0	0	0	0
Cinnamon	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total (100g)														
Per Serving Unrounded (40g)														
Rounded Values (40g)														
Percent DV														
Rounded Percent DV														

Total Fat	Total Carbs	Protein	Total	Total Rounded
Total Calories (unrounded)				

Extension Question

	Total Sugar (g)	Added Sugar (g)
New Natural Flavor	6	6
Total (100g)		
Per Serving Unrounded (40g)		
Rounded Values (40g)		
Percent DV		
Rounded Percent DV		